Checklists at the Speed of Prefill:

Local, Calibrated Clinical Eligibility Screening with Open Decision Models

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Abstract

Objective. Much of standard-of-care qualification means applying a published checklist to a free-text chart. We asked whether open *decision models*, a new class of models that answer typed yes/no questions with calibrated probabilities from a single prefill pass, can do this on a single laptop. We also asked which practical steps make them safe to use.

Methods. We encoded the USPSTF 2021 lung-cancer and 2019 abdominal aortic aneurysm screening recommendations as one question per criterion, and combined the answers into an auditable conjunctive eligibility score. We evaluated ten local systems on 900 synthetic notes from held-out templates with exact per-criterion labels, 40 hand-written notes, and 600 eligible-versus-excluded patient–trial pairs from TREC Clinical Trials 2021. Nine were decision models from four model families released between September and November 2026, and one was a generative large language model. We tested a deterministic hook that states smoking arithmetic, an 85-minute on-laptop fine-tune from synthetic notes, and automation thresholds chosen for a fixed error budget.

Results. Zero-shot, the newer open decision models clearly outperformed earlier ones on the lung checklist: conjunctive AUROC on template notes was 0.977 for Kev-27B and 0.976 for Clef-flash 9B, against 0.877 for Kev-4B. Laya was near chance (0.55–0.75). Criteria that require arithmetic were the main source of error for every model. A deterministic hook that states pack-years and years since quitting raised hand-written-note AUROC to 0.997–1.000 for the strongest models; with it, Kev-27B could auto-decide 95% of hand-written notes at $\leq$5% error. An 85-minute on-laptop fine-tune lifted Laya from 0.55 to 0.90 on hand-written notes (0.96 with the hook; 124 ms per note, about 44$\times$ faster than Kev-27B). AAA screening, whose criteria are stated explicitly, was essentially solved by every decision model from 4B upward. On trial prescreening, decision models of 4B–27B reached AUROC 0.82–0.90, above an 8B generative baseline (0.80); only Kev-27B could auto-decide a substantial share of pairs (32% at $\leq$5% error). Automation thresholds measured on synthetic notes overstated safe coverage on hand-written notes by up to 2.6-fold.

Conclusions. Open decision models can screen checklists locally, with no note leaving the machine, provided that the arithmetic is made explicit, the model is specialised or chosen on the target task, and the automation threshold is measured on the practitioner's own notes. We release the pipeline as a step-by-step guide (Appendix A), together with a local screening app, data generators and every prediction.

1 Introduction

Clinical qualification work often reduces to a checklist over a chart. Examples are the USPSTF recommendation that adults aged 50–80 with a 20 pack-year smoking history be offered annual low-dose CT [Krist et al., 2021], or a trial's inclusion and exclusion criteria. The facts these checklists

need often exist only in narrative text, such as "1 ppd since he was 18" or "quit after his MI in 2019". Rule engines over structured fields miss them. Generative large language models (LLMs) can read them [Jin et al., 2024, Wornow et al., 2025], but they are hard to run on premises. Their free-text output must be parsed, and their confidence is not calibrated by default.

A new class of models answers *typed questions* about a text in a single prefill pass: *decision models*, also called System One models. TypeSafe introduced the category on 15 September 2026 with Jev and its `/v1/systemone` API [TypeSafe, 2026, TypeSafe AI, 2026]. Within seven weeks it had become crowded:

- OpenAI's Decisions API reached general availability [OpenAI, 2026].
- Microsoft, Liquid AI and others launched hosted models [Microsoft, 2026, Liquid AI, 2026].
- Perplexity, Cloudflare, H2O.ai, IBM and the vLLM project released open weights [Perplexity AI, 2026, Cloudflare, 2026, H2O.ai, 2026, IBM, 2026, vLLM Semantic Router, 2026].
- Chrome prototyped an in-browser Decisions API [Google Chrome Built-in AI team, 2026], and llama.cpp added a native server for the API [Nguyen et al., 2026].
- Independent leaderboards appeared [Passos and Hugging Face, 2026, Standhartinger, 2026, Artificial Analysis, 2026], together with a first wave of clinical evaluations (§2).

For a checklist the interface has the right shape: one question per criterion, one probability for each, and a principled basis for deciding which cases are safe to automate.

The field now offers dozens of models but no answer to the question a clinical team actually faces: *how do we run our own checklist safely, on our own hardware, without sending notes anywhere?* This paper addresses that question. Our contributions are:

1. a screening pipeline that turns any public checklist into per-criterion decision questions with an auditable conjunctive score, written up as a step-by-step practitioner guide (Appendix A);
2. a controlled benchmark with exact per-criterion labels, held-out templates and hand-written notes, on which ten local systems are compared;
3. evidence that two inexpensive steps, a deterministic arithmetic hook and a short on-laptop fine-tune from synthetic notes, change which models are usable;
4. a measured demonstration that automation thresholds do not transfer from synthetic to hand-written notes;
5. an open-source release, including a local screening app, in which no note leaves the machine.

2 Background and related work

2.1 Decision models

A decision model reads a *state* (a note, a JSON record) and a set of questions whose answers are declared up front, and returns a probability for every answer instead of generated text. The System One API defines three question types: `noul` (yes/no), `choice` and `score` [TypeSafe, 2026]. The two open designs we study first span the range of architectures. **Laya** [Mukkunnoth, 2026] packs a question, its options (each introduced by a `[MASK]` marker) and the state into one sequence for a bidirectional encoder (ModernBERT-large, 421M parameters; Warner et al., 2024), and a two-layer head scores each marker. It is trained ("RLCD") with a GRPO-style policy gradient [Shao et al., 2024] on strictly proper scoring rules [Gneiting and Raftery, 2007] plus soft cross-entropy, and is then temperature-scaled [Guo et al., 2017]. **Kev** [Palmer, 2026] freezes a Qwen3.5 base and trains a rank-16 LoRA [Hu et al., 2022] and a pointer head, which scores each option's closing token against a final `<decide>` token. The note is prefilled once, and each question reuses the cache. Most open entrants since then (Table 1) follow Kev's pattern: a 4B–31B Qwen or Gemma decoder

Table 1: The decision-model landscape on 4 November 2026 (selection). "S1" = TypeSafe's `/v1/systemone` request shape. "Mac" = runs on a 64 GB Apple-silicon laptop. DI = independent Decision Index 0.4.1 Full score [Passos and Hugging Face, 2026] (chance-corrected, higher is better; "–" = not listed). Data-governance column quotes only what the provider states publicly; "none stated" means we found no statement, not that none exists. † = evaluated in this paper.

Model	Organisation (date)	Weights	Size / base	Interface	Clinical data governance (stated)	DI
Jev 1.13 / 1.14	TypeSafe (15 Sep / 14 Oct)	hosted	undisclosed	S1 (origin)	ZDR for enterprise; no HIPAA/BAA found	60.2
Decisions API (gpt-6-luna)	OpenAI (GA 22 Oct)	hosted	undisclosed	own (`/v1/decisions`)	ZDR and HIPAA for eligible customers	–
Microsoft-Decision-1	Microsoft (9 Oct)	hosted	Qwen3.5-9B	S1	data zones; HIPAA not stated	–
d1 / d1-ultra	Liquid AI (29 Sep)	hosted	undisclosed	S1	none stated	–
pplx-decider v1.2 (27B)	Perplexity (1–15 Oct)	open, Apache-2.0	Qwen3.8-27B	S1-style	ZDR by agreement (hosted)	68.0
Clef / Clef-flash	Cloudflare (1 Oct)	open, Apache-2.0	27B / 9B (Qwen)	S1	Workers AI: zero retention	54.9 / 44.4
H2O-Lightning-4B	H2O.ai (2 Oct)	open, Apache-2.0	Qwen3.5-4B	S1 (llama.cpp)	local	–
Decision 2.0 (Nox-4B … Vega-27B)	vLLM Semantic Router (28 Sep)	open, Apache-2.0	Qwen3.5/3.8	S1 (llama.cpp)	local	–
Granite Switch 4.1	IBM (14 Oct)	open, Apache-2.0	8B / 30B	chat + S1	local	–
Decisions API (`DecisionModel`)	Google Chrome (prototype, Oct)	on-device	undisclosed	own web API	"inputs not sent over the network"	–
Kev 1.0 (0.8B–27B)	J. Palmer (Sep–Oct)	open, Apache-2.0	Qwen3.5/3.8 + LoRA	S1	local	59.6 (27B), 40.9 (9B), 34.9 (4B)
Laya	ConvAI Innovations (18 Sep)	open, Apache-2.0	ModernBERT 421M	S1	local	9.2
MedJev	Lee et al. (26 Oct)	open, CC-BY-NC	9B	S1 (shim)	local	–
CS1-4B	Zhang et al. / Mahmood lab (19 Oct)	not released	Decision 2.0 Nox-4B	S1	trained on 3.6M patients	–

with a readout head, one temperature, and the System One request shape. Because they share that request shape, one client can drive almost all of them.

2.2 Independent benchmarks

On the Hugging Face Decision Index [Passos and Hugging Face, 2026], the best open models (71–72) now score above Jev 1.14 (60.2). Kev-27B (59.6) is level with Jev; Kev-4B scores 34.9, and Laya scores 9.2 and ranks 251st of 265. JevBench [Standhartinger, 2026] and the Artificial Analysis Decision Index [Artificial Analysis, 2026] give the same broad ordering, and find that decision models remain overconfident on knowledge questions. These boards measure general-purpose decisions, not clinical checklists. Laya's author has publicly claimed priority over Jev's architecture [Ferreira, 2026]; we cite both and take no position.

2.3 Clinical eligibility from text

Cohort selection from narrative records was benchmarked in the n2c2 2018 shared task [Stubbs et al., 2019]. Criteria2Query translates free-text criteria into database queries [Yuan et al., 2019], and the TREC Clinical Trials track provides graded patient–trial judgements [Roberts et al., 2021]. LLM-based matchers achieve strong criterion-level performance with large generative models, usually hosted [Jin et al., 2024, Wornow et al., 2025].

2.4 Clinical evaluations of decision models

Most early studies tested hosted Jev or large open decision models on medical knowledge and diagnosis. Accuracy is reasonable on examination-style questions, but calibration degrades on patient-level prediction and complex cases [Du et al., 2026, Madrid-García et al., 2026, You et al., 2026, Deobhakta et al., 2026]. Laya was "near chance" in rheumatology [Madrid-García et al., 2026]. Three studies are closest to ours:

- He et al. [2026] evaluated hosted Jev under zero data retention on 2,048 TrialGPT and MIMIC patient–criterion pairs. It matched a schema-constrained LLM on macro-F1, flagged insufficient information more reliably, and cost 53–74% less.
- Zhang et al. [2026] trained CS1-4B from Decision 2.0 Nox-4B on 12.4M questions from 3.6M patients. It outperformed open models of up to 27B on 13 of 14 tasks and quadrupled the number of eligible patients in the top 20 for trial screening. Its weights are not public.

- Lee et al. [2026] released MedJev and reported a patient-level ECE of 0.286, reduced to 0.007 by recalibration on 50–200 labelled examples.

These works evaluate models. We provide a procedure a practitioner can follow, which needs neither hospital-scale training data nor a data-processing agreement, and we test each of its steps.

2.5 Selective prediction

A calibrated probability allows a system to abstain. We use the risk–coverage framework [Geifman and El-Yaniv, 2017]: cases are ranked by confidence, and we report the largest share that can be auto-decided while the error rate among them stays within a budget.

3 Methods

3.1 Screening pipeline

The pipeline has seven steps (full guide with code in Appendix A):

1. encode each criterion as a `noul` question with the answer that eligibility requires;
2. choose a local decision model and serve it behind the System One API;
3. optionally append deterministic arithmetic facts to the note;
4. combine the criterion probabilities into a conjunctive eligibility score;
5. choose an automation threshold on labelled notes for a fixed error budget;
6. if coverage is too low, fine-tune locally on synthetic notes;
7. deploy with clinician review of the cases below the threshold.

The sections below specify each component as used in the experiments.

3.2 Checklist encoding and the conjunctive score

The **lung-cancer screening** recommendation (USPSTF 2021, Grade B; Krist et al., 2021) yields four criteria: age 50–80; ≥20 pack-years; currently smokes or quit ≤15 years ago; and no health problem that substantially limits life expectancy or the ability or willingness to undergo curative lung surgery. The last criterion must be answered "no". The **AAA screening** recommendation (USPSTF 2019, Grade B; Owens et al., 2019) yields three: male, age 65–75, ever smoked. Each criterion is a `noul` question stating its thresholds, inclusivity and units (Appendix B). For criteria c_i with required answers $r_i \in \{\text{yes}, \text{no}\}$ and model probabilities $p_i = P(c_i = \text{yes} \mid \text{note})$, the conjunctive score is

$$P(\text{eligible}) = \prod_i \left[p_i \, \mathbf{1}(r_i = \text{yes}) + (1 - p_i) \, \mathbf{1}(r_i = \text{no}) \right]. \tag{1}$$

This treats the criteria as conditionally independent given the note, which is a heuristic. In return, every negative decision names the criterion responsible. As a comparison we also ask a single *direct* question containing the whole recommendation.

3.3 Computed-facts hook

Decision models are weakest where a checklist needs arithmetic: packs per day × years, cigarettes converted to packs, or a quit year compared with the visit year. A deterministic pre-processor (`clinical_system1/facts.py`) finds these spans with generic regular expressions and appends the computed values, for example `[Computed facts: about 40 pack-years (1 packs/day x 40 years); age 58.]`. The model still makes every judgement. When no pattern matches, nothing is appended. This mirrors Kev's `KEV_DATE_FACTS` option for day counts.

3.4 Data

E1: synthetic vignettes. Primary-care notes are rendered from structured seeds (age, sex, smoking status, packs per day, years smoked, years since quitting, life-limiting condition, refusal of surgery), so every criterion label is exact. Smoking history is phrased as packs/day × years, cigarettes/day × years, or a stated pack-year figure; quitting is phrased as "N years ago" or "in YYYY" with the visit year given. Distractors include household smoking, secondhand exposure, a family history of lung cancer or AAA, and benign comorbidities. Life-limiting conditions are:

- hospice-enrolled metastatic cancer;
- NYHA IV heart failure;
- oxygen-dependent end-stage COPD;
- advanced dementia;
- ESRD with dialysis declined;
- explicit refusal of surgery.

Values at each threshold are oversampled. Template family A (bullet and short narrative formats; 3,000 lung and 2,000 AAA notes) is used only for fine-tuning and calibration. Family B (problem-list and long-narrative formats; 500 lung and 400 AAA notes, 40% eligible) is used only for testing.

E1b: hand-written notes. We wrote 40 free-text notes by hand with clinical abbreviations ("ppd", "pk-yr", "s/p"), indirect dates ("quit after his MI in 2019") and boundary cases (e.g. 10 cigarettes/day for 38 years = 19 pack-years). 13 are lung-eligible and 10 AAA-eligible.

E2: trial prescreening. From TREC Clinical Trials 2021 [Roberts et al., 2021] we sampled 600 judged patient–trial pairs, 300 eligible and 300 excluded, stratified over the 75 topics. Eligibility text was taken from the ClinicalTrials.gov v2 API, with each criteria block capped at 6,000 characters. We ask a direct eligibility question plus "meets the inclusion criteria?" and "meets any exclusion criterion?", and score $P(\text{inc})(1 - P(\text{exc}))$.

3.5 Models

All systems run locally on one Apple M1 Max laptop (64 GB unified memory). Table 2 lists them. Kev-0.8B, Kev-4B and Kev-9B are served by Kev's own MLX server; Kev-27B, H2O-Lightning-4B and Clef-flash by llama.cpp's native `/v1/systemone` endpoint (build 11960). Two settings were needed on this hardware: `GGML_METAL_BF16_DISABLE=1`, because the M1 GPU lacks bfloat16, and, for Clef-flash, a physical batch at least as long as the input (`-b16384-ub16384`). All decision models are queried through the same System One client with their shipped calibration. We do not apply H2O's optional "yes/no floor", which changes reported probabilities. The generative baseline, Gemma 4 (about 8B, 4-bit, via Ollama), is asked one question at a time; $P(\text{yes})$ is the softmax over its first-token log-probabilities for "Yes" and "No". For time reasons Gemma 4 was run on the first 250 notes of each E1 template test set and a balanced 300-pair E2 subset. Kev-27B (about 5 s per note) was run on the first 250 notes of each E1 template test set, all hand-written notes, and a balanced 150-pair E2 subset. We tried to include further medical and open models but excluded them for reproducibility reasons:

- MedJev [Lee, 2026] and MedDecider (non-commercial) require bespoke client runtimes from their model repositories;
- a community GGUF of pplx-decider was incompatible with stock llama.cpp;
- Decision 2.0 Nox-4B, the base of CS1-4B, needs a llama.cpp build newer than the packaged release.

Table 2: Systems evaluated. All run on one Apple M1 Max (64 GB). FT = fine-tuned in this work.

System	Parameters / precision	Runtime	Released
Laya EN, Laya multilingual (zero-shot)	421M / 322M, fp32	PyTorch MPS	Sep 2026
Laya EN + RLCD FT (ours)	421M, fp32	PyTorch MPS	this work
Kev-0.8B, Kev-4B, Kev-9B v2	LoRA on Qwen3.5, bf16	Kev MLX server	Sep–Oct 2026
Kev-27B v2	27B, Q4_K_M GGUF	llama.cpp	Oct 2026
H2O-Lightning-4B	4B, Q8_0 GGUF	llama.cpp	Oct 2026
Cloudflare Clef-flash	9B, Q8_0 GGUF	llama.cpp	Oct 2026
Gemma 4 (generative reference)	∼8B, 4-bit	Ollama	2026

3.6 Local fine-tuning

We ported Laya's released two-GPU RLCD notebook to a single Apple GPU, keeping its hyperparameters:

- AdamW with learning rates 2.5×10^{-5} (encoder) and 10^{-4} (head), cosine schedule;
- micro-batch 8 with 4-step accumulation;
- G=4 noisy samples, with σ annealed from 0.4 to 0.1;
- a reward combining log, spherical (w=0.75) and ranked-probability (w=1) scores, plus soft cross-entropy.

Laya's maintainers have since published their own Apple-silicon script. We trained for 2 epochs on 22,000 criterion and direct questions from family A, then fitted a `noul` temperature by LBFGS on 1,000 held-out family-A questions. Family-B notes and hand-written notes were never seen in training or calibration.

3.7 Automation thresholds and metrics

For a score P, confidence is $\max(P, 1-P)$ and the decision is $P \geq 0.5$. Notes are ranked by confidence. *Coverage at budget b* is the largest fraction of notes that can be auto-decided, most confident first, with an error rate $\leq b$ among them; the corresponding confidence is the automation threshold τ. We also report AUROC with 95% bootstrap confidence intervals (1,000 resamples), accuracy at 0.5, balanced accuracy, Brier score, 10-bin expected calibration error (ECE), and the fraction of notes on which every criterion is answered correctly ("all criteria right"). Latency is the median wall-clock time per note for the full five-question lung checklist, measured with nothing else running on the GPU.

4 Results

4.1 Zero-shot performance of decision models

Tables 3–7 and Figure 1 summarise the checklist results. **The newer open decision models are markedly stronger zero-shot.** On the lung template notes the conjunctive AUROC is 0.976 for Clef-flash (9B) and 0.919 for H2O-Lightning-4B, against 0.877 for Kev-4B and 0.890 for Kev-9B. The gap is clearest in the audit trail: every criterion is right on 80% of notes for Clef-flash, 51% for H2O-Lightning-4B, and 40% and 17% for Kev-4B and Kev-9B. Kev-9B mislabels the life-limiting criterion on most notes. Kev-27B (250 notes, 4-bit) reaches a conjunctive AUROC of 0.977 with every criterion right on 73% of notes. AAA, whose criteria are stated explicitly, is essentially solved by every Kev model from 4B upward and by both new models (AUROC $\geq$0.998). **Laya zero-shot**

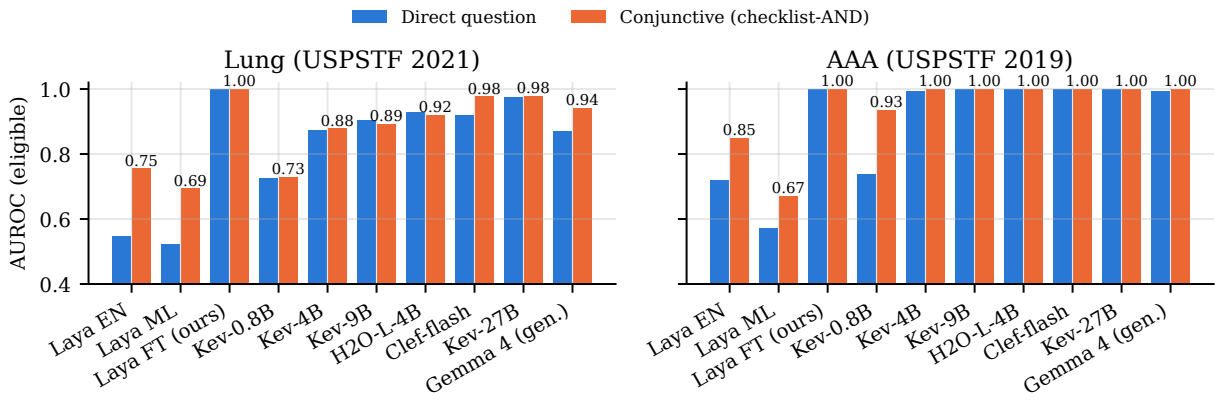


Figure 1: Eligibility AUROC on held-out-template notes: one direct question (blue) versus the conjunctive score (orange).

Table 3: USPSTF 2021 lung cancer screening eligibility on 500 held-out-template notes (40% eligible). Coverage = share of notes auto-decided at the given error budget. Bold = best.

	Direct	Checklist-AND					All crit.	
Model	AUROC	AUROC	Acc	Brier↓	Cov@5%	Cov@10%	correct	n
Laya EN (0-shot)	0.547	0.754	0.600	0.369	0.130	0.168	0.082	500
Laya ML (0-shot)	0.523	0.693	0.608	0.284	0.000	0.130	0.242	500
Laya EN + RLCD FT (ours)	**0.998**	**0.999**	**0.992**	**0.008**	**1.000**	**1.000**	**0.972**	500
Kev-0.8B	0.726	0.728	0.602	0.260	0.268	0.316	0.288	500
Kev-4B	0.872	0.877	0.648	0.244	0.424	0.492	0.402	500
Kev-9B v2	0.903	0.890	0.602	0.289	0.390	0.466	0.166	500
H2O-Lightning-4B	0.929	0.919	0.632	0.219	0.370	0.568	0.508	500
Cloudflare Clef-flash 9B	0.918	0.976	0.822	0.105	0.664	0.822	0.800	500
Kev-27B v2 (Q4)	0.973	0.977	0.912	0.054	0.916	**1.000**	0.732	250
Gemma 4 (generative)	0.870	0.940	0.888	0.085	0.776	0.924	0.712	250

is not usable for this task. Lung AUROC is 0.55–0.75, all four criteria are right on only 8% of notes, and it misjudges even the explicit age criterion (44% accuracy). This matches its position near the bottom of the general leaderboards and upstream's warning that the `noul` head can follow its label text instead of the state.

4.2 Which criteria fail

Arithmetic criteria are the bottleneck for every zero-shot model (Table 5). For Kev-4B, age is 98% correct but pack-years only 77% and recency 68%; Kev-9B is similar on recency (69%). The newer models handle arithmetic better but not perfectly: pack-years 88% and 91%, and recency 80% and 91%, for H2O-Lightning-4B and Clef-flash respectively. Kev-27B scores 88% on pack-years and 92% on recency.

4.3 Effect of the computed-facts hook

The hook raises pack-year accuracy for every model (Table 6). Kev-4B goes from 0.77 to 0.98 on template notes and from 0.82 to 1.00 on hand-written notes; Gemma 4 goes from 0.84 to 0.99.

Table 4: Lung screening on 40 hand-written free-text notes (13 eligible).

	Direct	Checklist-AND					All crit.	
Model	AUROC	AUROC	Acc	Brier↓	Cov@5%	Cov@10%	correct	n
Laya EN (0-shot)	0.462	0.553	0.675	0.302	0.075	0.075	0.100	40
Laya ML (0-shot)	0.590	0.293	0.675	0.302	0.025	0.025	0.175	40
Laya EN + RLCD FT (ours)	**0.913**	**0.903**	**0.875**	**0.122**	0.375	**0.750**	**0.850**	40
Kev-0.8B	0.821	0.553	0.675	0.238	0.100	0.100	0.375	40
Kev-4B	0.869	0.732	0.650	0.254	0.275	0.325	0.500	40
Kev-9B v2	0.880	0.752	0.675	0.259	0.250	0.425	0.225	40
H2O-Lightning-4B	0.846	0.832	0.675	0.205	0.550	0.575	0.575	40
Cloudflare Clef-flash 9B	0.889	0.883	0.700	0.166	0.550	0.600	0.700	40
Kev-27B v2 (Q4)	0.895	0.897	0.775	0.152	**0.575**	0.625	0.725	40
Gemma 4 (generative)	0.838	0.840	0.800	0.163	0.250	0.600	0.650	40

Table 5: Per-criterion accuracy on the lung-screening test set (500 notes, template family B).

Model	Age 50–80	≥20 pack-yr	Recency	Limiting
Laya EN (0-shot)	0.444	0.592	0.486	0.562
Laya ML (0-shot)	0.708	0.634	0.520	0.878
Laya EN + RLCD FT (ours)	1.000	0.992	0.980	1.000
Kev-0.8B	0.722	0.686	0.622	0.922
Kev-4B	0.982	0.774	0.682	0.830
Kev-9B v2	0.934	0.886	0.686	0.298
H2O-Lightning-4B	0.940	0.876	0.804	0.774
Cloudflare Clef-flash 9B	1.000	0.906	0.912	0.970
Kev-27B v2 (Q4)	0.956	0.876	0.920	0.964
Gemma 4 (generative)	0.968	0.840	0.888	0.976

For the fine-tuned Laya on hand-written notes, AUROC rises from 0.903 to 0.957 and coverage at a 5% error budget from 38% to 85%. Kev's *recency* accuracy does not change (0.68) even when "quit about 16 years before the visit" is stated, which indicates that it does not reliably compare a stated number with the 15-year cut-off inside a disjunctive question. The newer models benefit as much. With the hook, Clef-flash reaches pack-year and recency accuracies of 1.00 and 0.97 on hand-written notes, and AUROC 1.000 (direct) and 0.997 (conjunctive). H2O-Lightning-4B reaches 1.00 and 1.00, with AUROC 0.989 (direct). Kev-27B reaches AUROC 1.000 (direct) and 0.997 (conjunctive), and the highest safe-automation share in the study on hand-written notes (95% at a 5% error budget with the conjunctive score; 100% with the direct question). The hook can also hurt. For H2O-Lightning-4B on template notes, accuracy on the unrelated life-limiting criterion fell from 0.77 to 0.59 when facts were appended, so any hook must be validated on every criterion, not just the ones it targets. The hook computed pack-years for all 433 template smokers and for 34 of the 36 hand-written smokers. The two misses were "from her 20s until 2005" and "a few cigarettes on weekends".

4.4 Direct question versus conjunctive score

On template notes the conjunctive score is at least as good as the direct question for every system (e.g. Gemma 4 lung 0.870→0.940; Kev-0.8B AAA 0.736→0.934), and its audit trail is fully correct on 97% of notes for the fine-tuned Laya. It is not uniformly better. For zero-shot Kev on hand-

Table 6: Step 3 ablation: the computed-facts hook on the lung checklist (checklist-AND). Δ = with minus without.

Notes	Model	AUROC	Pack-yr acc	Recency acc	Cov@5%	All crit.
Template	Kev-0.8B	0.81 (+0.08)	0.84 (+0.16)	0.63 (+0.01)	0.35 (+0.08)	0.34 (+0.06)
Template	Kev-4B	0.94 (+0.06)	0.98 (+0.20)	0.68 (+0.00)	0.52 (+0.10)	0.44 (+0.04)
Template	Kev-9B v2	0.97 (+0.08)	1.00 (+0.11)	0.68 (-0.00)	0.55 (+0.16)	0.19 (+0.03)
Template	H2O-Lightning-4B	0.96 (+0.05)	0.99 (+0.12)	0.92 (+0.11)	0.42 (+0.05)	0.52 (+0.01)
Template	Cloudflare Clef-flash 9B	1.00 (+0.02)	1.00 (+0.09)	0.98 (+0.07)	0.96 (+0.30)	0.92 (+0.12)
Template	Kev-27B v2 (Q4)	1.00 (+0.02)	1.00 (+0.12)	0.98 (+0.06)	1.00 (+0.08)	0.88 (+0.15)
Template	Laya EN (0-shot)	0.70 (-0.06)	0.75 (+0.16)	0.70 (+0.21)	0.18 (+0.05)	0.11 (+0.03)
Template	Laya EN + RLCD FT (ours)	0.99 (-0.00)	1.00 (+0.01)	0.94 (-0.04)	1.00 (+0.00)	0.93 (-0.04)
Template	Gemma 4 (generative)	1.00 (+0.06)	0.99 (+0.15)	0.92 (+0.03)	1.00 (+0.22)	0.88 (+0.16)
Hand-written	Kev-0.8B	0.64 (+0.09)	0.85 (+0.05)	0.70 (-0.03)	0.15 (+0.05)	0.42 (+0.05)
Hand-written	Kev-4B	0.76 (+0.03)	1.00 (+0.18)	0.68 (+0.00)	0.25 (-0.03)	0.55 (+0.05)
Hand-written	Kev-9B v2	0.87 (+0.11)	1.00 (+0.12)	0.68 (+0.00)	0.50 (+0.25)	0.25 (+0.02)
Hand-written	H2O-Lightning-4B	0.95 (+0.12)	1.00 (+0.20)	1.00 (+0.18)	0.65 (+0.10)	0.72 (+0.15)
Hand-written	Cloudflare Clef-flash 9B	1.00 (+0.11)	1.00 (+0.18)	0.97 (+0.07)	0.82 (+0.27)	0.90 (+0.20)
Hand-written	Kev-27B v2 (Q4)	1.00 (+0.10)	1.00 (+0.10)	0.97 (+0.15)	0.95 (+0.38)	0.90 (+0.18)
Hand-written	Laya EN (0-shot)	0.76 (+0.21)	0.78 (+0.22)	0.62 (+0.10)	0.40 (+0.33)	0.20 (+0.10)
Hand-written	Laya EN + RLCD FT (ours)	0.96 (+0.05)	0.95 (+0.10)	0.93 (-0.05)	0.85 (+0.47)	0.88 (+0.03)
Hand-written	Gemma 4 (generative)	0.99 (+0.15)	0.97 (+0.12)	0.82 (+0.05)	0.80 (+0.55)	0.78 (+0.12)

written lung notes the direct question wins (Kev-4B 0.869 vs 0.732; with facts 0.966 vs 0.764), because a single miscalibrated criterion (recency) pulls the product down and sensitivity falls to 0.15. The same pattern, high AUROC but low sensitivity at a 0.5 cut-off, appears for most zero-shot models, because a product of four moderately confident "yes" answers rarely exceeds 0.5. This is a further reason to set the operating point on labelled notes rather than at 0.5.

4.5 Local fine-tuning

Fine-tuning took 85 minutes and fitted a `noul` temperature of 3.5 (0.996 accuracy on the held-out calibration slice). On the unseen family-B templates the fine-tuned Laya reaches lung AUROC 0.999 and AAA 1.000, with Brier 0.008 and every criterion right on 97% of notes. Because the two template families share phrase vocabulary, the hand-written notes are the stricter test. There it scores lung AUROC 0.903 (0.957 with facts), with every criterion right on 85% (87.5%) of notes, compared with 0.553 and 10% before fine-tuning. On hand-written notes the fine-tuned 421M encoder is therefore competitive with the strongest zero-shot models in the study (conjunctive AUROC without/with facts: Kev-27B 0.897/0.997; Clef-flash 0.883/0.997; H2O-Lightning-4B 0.832/0.949), at between a tenth and a fortieth of their latency, although the 27B model automates more notes safely (95% vs 85%). With only 40 notes the confidence intervals of the top systems overlap (Table 4).

4.6 Selective automation and threshold transfer

Coverage at a fixed error budget is the most practical quantity here and also the most fragile (Figure 2). The fine-tuned Laya automates 100% of template notes at ≤5% error with τ=0.98, but only 38% of hand-written notes (85% with the facts hook). Kev-4B automates 42% of template notes but 28% of hand-written ones. The stronger new models degrade less: Kev-27B with facts falls from 100% to 95% and Clef-flash with facts from 96% to 82% (conjunctive score), and H2O-Lightning-4B with facts and the direct question from 100% to 90%. A threshold measured on

Table 7: USPSTF 2019 AAA screening eligibility on 400 held-out-template notes (40% eligible).

	Direct	Checklist-AND					All crit.	
Model	AUROC	AUROC	Acc	Brier↓	Cov@5%	Cov@10%	correct	n
Laya EN (0-shot)	0.720	0.849	0.600	0.300	0.417	0.468	0.285	400
Laya ML (0-shot)	0.571	0.669	0.630	0.229	0.007	0.028	0.355	400
Laya EN + RLCD FT (ours)	**1.000**	**1.000**	**1.000**	0.000	**1.000**	**1.000**	**1.000**	400
Kev-0.8B	0.736	0.934	0.848	0.131	0.527	0.777	0.613	400
Kev-4B	0.993	1.000	0.988	0.017	**1.000**	**1.000**	0.965	400
Kev-9B v2	0.998	0.998	0.983	0.018	**1.000**	**1.000**	0.965	400
H2O-Lightning-4B	1.000	**1.000**	0.988	0.026	**1.000**	**1.000**	0.990	400
Cloudflare Clef-flash 9B	**1.000**	**1.000**	0.998	0.021	**1.000**	**1.000**	0.993	400
Kev-27B v2 (Q4)	**1.000**	**1.000**	**1.000**	0.002	**1.000**	**1.000**	**1.000**	250
Gemma 4 (generative)	0.994	**1.000**	**1.000**	**0.000**	**1.000**	**1.000**	0.996	250

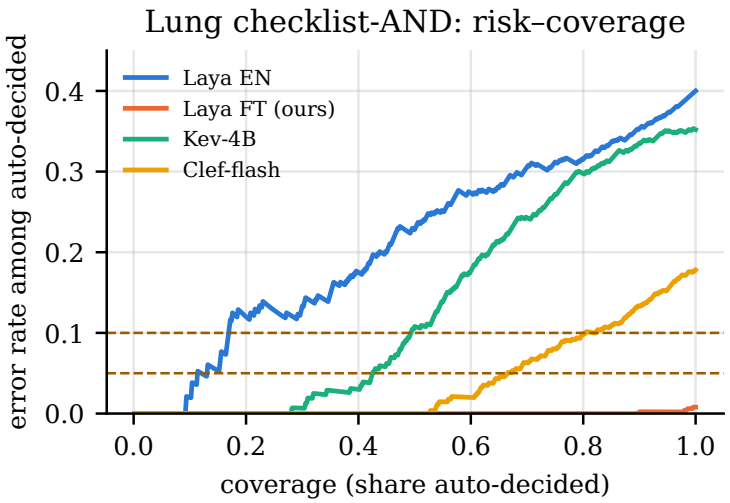


Figure 2: Risk–coverage for the lung conjunctive score on held-out-template notes. A curve that stays under the dashed error budget longer automates more notes.

synthetic notes would therefore overstate safe coverage on real text, by a factor of 2.6 for the fine-tuned model.

4.7 Trial prescreening

Distinguishing eligible from excluded patients for a topically relevant trial is harder (Table 8). Kev-4B reaches AUROC 0.820 [0.78, 0.85] directly and 0.839 [0.81, 0.87] with the inclusion×(1−exclusion) decomposition. Kev-9B is similar (0.833 / 0.836) and Gemma 4 slightly lower (0.803 / 0.797). The newer models are slightly better: H2O-Lightning-4B reaches 0.847 [0.82, 0.88] and Clef-flash 0.836 [0.80, 0.87]; Kev-27B reaches 0.903 [0.86, 0.94] on a balanced 150-pair subset. The differences among the 4B–9B decision models are within their confidence intervals; all 4B–27B decision models score above the generative baseline (0.803). Safe automation is limited: at a 5% error budget the 4B–9B models auto-decide at most 13% of pairs (best of the two scores), and the generative baseline 12%. Kev-27B reaches 32% with the direct question. In this setting trial prescreening is mainly a triage aid, with the largest model beginning to make partial automation plausible.

Table 8: TREC Clinical Trials 2021, eligible vs. excluded (balanced; 95% bootstrap CI for AUROC).

	Direct		Inc.×(1−Exc.)		
Model	AUROC	Bal. acc	AUROC	Bal. acc	n
Laya ML (8k ctx)	0.514 [0.47, 0.56]	0.503	0.481 [0.43, 0.53]	0.500	600
Kev-0.8B	0.633 [0.59, 0.68]	0.515	0.627 [0.58, 0.67]	0.587	600
Kev-4B	0.820 [0.78, 0.85]	0.732	0.839 [0.81, 0.87]	0.652	600
Kev-9B v2	0.833 [0.80, 0.86]	0.732	0.836 [0.80, 0.87]	0.707	600
H2O-Lightning-4B	0.847 [0.82, 0.88]	0.752	0.846 [0.82, 0.88]	0.742	600
Cloudflare Clef-flash 9B	0.836 [0.80, 0.87]	0.688	0.836 [0.80, 0.87]	0.698	600
Kev-27B v2 (Q4)	0.903 [0.86, 0.94]	0.820	0.894 [0.84, 0.94]	0.807	150
Gemma 4 (generative)	0.803 [0.75, 0.85]	0.560	0.797 [0.74, 0.84]	0.623	300

Table 9: Latency for the full 5-question lung checklist per note on one Apple M1 Max, uncontended (50 notes after 3 warm-up calls).

System	Params	p50 ms	p90 ms	notes/s
Laya EN, Apple GPU (MPS)	421M	126	136	7.96
Laya EN, CPU only	421M	506	571	1.93
Laya FT (ours), MPS	421M	124	136	8.04
Kev-0.8B, MLX bf16	0.8B	134	137	7.68
Kev-4B, MLX bf16	4B	694	703	1.49
Gemma 4, Ollama	∼8B	1250	1373	0.79
Kev-9B v2, MLX bf16*	9B	1575	1730	0.64
Kev-27B v2, llama.cpp Q4*	27B	5354	5841	0.19
H2O-Lightning-4B, llama.cpp Q8*	4B	1053	1135	0.95
Clef-flash, llama.cpp Q8*	9B	1468	1546	0.68

* median over all 500 lung test notes from the benchmark run itself (uncontended, sequential).

4.8 Latency

For the full five-question lung checklist on one M1 Max (Table 9), the fine-tuned Laya takes 124 ms per note on the GPU and 506 ms on the CPU alone. Kev-0.8B takes 134 ms, Kev-4B 694 ms and Gemma 4 1,250 ms. Through llama.cpp, H2O-Lightning-4B takes about 1.05 s per note and Clef-flash about 1.47 s; Kev-9B takes 1.58 s and Kev-27B (4-bit) about 5.4 s. At 124 ms per note, a single laptop screens about 29,000 notes per hour.

5 Discussion

5.1 Principal findings

Five findings matter most for practice.

1. The typed-question interface maps onto clinical checklists with no glue code and yields a per-criterion audit trail.
2. Open decision models improved sharply between September and November 2026. The strongest 4B–9B models released in October (H2O-Lightning-4B, Clef-flash) screen explicit criteria nearly perfectly zero-shot and are much better than earlier open models on criteria that need arithmetic.

3. Stating the arithmetic with a deterministic hook was the single most effective step for every model, and with it the best zero-shot open model reached AUROC 1.000 on hand-written lung notes. The hook can also degrade an unrelated criterion, so it must be validated on all of them.
4. A short on-laptop fine-tune turned the weakest model on the general leaderboards into one competitive with the strongest zero-shot models on this checklist, at a tenth to a fortieth of their latency, although the largest model still automated more notes safely. Specialisation from synthetic notes is a practical option when throughput or CPU-only deployment matters.
5. The automation threshold must be measured on the target notes. A threshold taken from synthetic data overstated safe coverage by up to 2.6-fold, less for stronger models (Kev-27B 100% vs 95%) but never zero.

5.2 Comparison with prior work

Our threshold finding agrees with Lee et al. [2026]: out-of-the-box calibration of decision models on patient-level tasks is poor, and a small labelled set (50–200 examples) is enough to repair it. CS1-4B [Zhang et al., 2026] shows what institution-scale clinical training achieves. Our fine-tuning result suggests that much of the benefit for a fixed checklist is available from synthetic notes on a laptop, although our task is far narrower than theirs. He et al. [2026] found hosted Jev competitive with a schema-constrained LLM on trial criteria; our TREC results show that open models running locally reach AUROC above 0.8 on a related task, although the two settings are not directly comparable. General leaderboards were a partial guide. The models that rank highest among those we could run (Kev-27B, Clef-flash) were also strong here, but the model ranked 251st of 265 on the Decision Index became competitive after fine-tuning.

5.3 Hosted or local deployment

Hosted decision APIs are now inexpensive ($0.01–0.10 per million input tokens), fast, and in some cases ahead on general leaderboards. For clinical notes the deciding factor is usually governance. Among the providers in Table 1, only OpenAI states HIPAA eligibility, and Cloudflare states zero data retention; the others offer retention terms under enterprise or written agreements, or make no statement. A fully local pipeline needs no agreement, and the step that matters most, measuring the threshold on one's own notes, is the same in either deployment.

5.4 Decision models and generative models

In our September experiments the generative baseline was the most accurate system on hand-written lung notes. That is no longer the case. With the facts hook, Gemma 4 reached AUROC 0.986 (conjunctive) and Clef-flash reached 0.997 (conjunctive) and 1.000 (direct), at similar latency on this laptop (1.25 s and 1.47 s per note). The decision model returns calibrated probabilities through a standard API with no output parsing; the generative model needs one decode per question and a custom log-probability readout. For high-throughput or CPU-only settings, the fine-tuned 421M encoder remains about ten times faster than either.

5.5 Limitations

- **Synthetic text.** E1 notes are template-generated. Disjoint template families and hand-written notes narrow the gap to real EHR text but do not close it.

- **Small hand-written set.** It contains 40 notes, written and labelled by the same author who wrote the facts hook, which may favour the hook.
- **TREC data.** TREC topics are synthetic and its relevance labels are noisy.
- **Independence assumption.** The conjunctive score treats criteria as independent given the note.
- **Local models only.** We evaluated open models that run locally, several in quantised form; hosted APIs are discussed only from published information. The generative baseline is a single 4-bit model without chain-of-thought.
- **Not validated for care.** Nothing here is validated for patient care. Deployment requires prospective evaluation on real charts, subgroup audits and clinician review of automated negatives.

6 Conclusion

Open decision models running on a laptop can screen standard-of-care checklists in a meaningful way, with no note leaving the machine. This requires that the checklist be broken into typed criteria, the arithmetic be made explicit, the model be chosen or specialised on the target task, and the automation threshold be measured on the practitioner's own notes. On our benchmark, the best zero-shot open model with an arithmetic hook reached AUROC 1.000 on hand-written lung notes, and a 421M encoder fine-tuned for 85 minutes on a laptop reached 0.957 at 124 ms per note. In both cases the share of notes that could be automated safely had to be measured on the target notes. The released guide, app and predictions allow others to repeat the process for their own checklists.

Data and code availability

Code, vignette generators, hand-written notes, per-item predictions, the screening app and HTML documentation are available at https://github.com/menonpg/clinical-system1 (Apache-2.0). TREC Clinical Trials 2021 topics and judgements are available from NIST, and trial text from ClinicalTrials.gov. Upstream models: https://github.com/NandhaKishorM/laya, https://github.com/jaredpalmer/kev, and the model cards cited in Table 1.

Use of AI assistance

Software, experiment orchestration and manuscript drafting were assisted by an AI coding assistant (Claude, Anthropic). The author designed the study, reviewed all code and results, and takes responsibility for the content.

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A Practitioner guide: screening your own checklist

This appendix walks through the pipeline using lung-cancer screening as the running example. Each step names the file in the release that implements it.

Step 1 — Write one yes/no question per criterion

Split the recommendation into atomic criteria, write each as a `noul` question, and record the answer that eligibility *requires* (`clinical_system1/checklists.py`):

```
LUNG = Checklist(key="lcs", title="Lung cancer screening (USPSTF 2021)", criteria=(
  Criterion("lcs_age", "Is the patient between 50 and 80 years old, inclusive?"),
  Criterion("lcs_pack_years", "Does the patient have a smoking history of at least"
            " 20 pack-years (packs per day x years smoked; 20 cigarettes = 1 pack)?"),
  Criterion("lcs_recency", "Does the patient currently smoke, or did they quit"
            " 15 or fewer years ago?"),
  Criterion("lcs_limiting", "Does the patient have a health problem that substantially"
            " limits life expectancy, or are they unwilling or unable to undergo"
            " curative lung surgery?", required=False)))
```

Rules we found useful:

- State thresholds and inclusivity explicitly.
- Define units in the question ("20 cigarettes = 1 pack").
- Never combine two criteria in one question.
- Phrase exclusions positively ("does X apply?") with `required=False`.
- Keep a direct question as a baseline.

Step 2 — Choose a model and serve it locally

Most open decision models can be served on a laptop with one command, and all speak the same request shape. Some need extra flags; for example Clef-flash needs `-b16384-ub16384` so that a whole note fits in one batch:

```
GGML_METAL_BF16_DISABLE=1 llama-server -m h2o-lightning-4b-Q8_0.gguf --no-mmproj -np 1 --port
    8031
cd upstream/kev && uv run --extra serve python -m kev.serve --run jaredpalmer/kev-4b --port 8009
```

```
from clinical_system1.backends import SystemOneBackend
model = SystemOneBackend("http://127.0.0.1:8031", "h2o-lightning-4b")
p = model.predict(note_text, LUNG.questions(include_direct=False))   # {criterion: P(yes)}
```

Leaderboard rank is only a partial guide for clinical checklists (§5). Shortlist two or three models that fit your hardware and licence (several medical models are non-commercial), and run Step 5 on your own notes for each. Use hosted APIs only under an agreement that covers protected health information.

Step 3 — Give the model the arithmetic

Append computed values with a deterministic hook (`clinical_system1/facts.py`), and write the equivalent for the units in your own checklist:

```
Visit 2026. 58M ... Smokes ~1 ppd since he was 18 ...
[Computed facts: about 40 pack-years (1 packs/day x 40 years); age 58.]
```

Step 4 — Combine the criteria

Compute Eq. (1) (`Checklist.aggregate`) and keep `Checklist.failed` as the audit trail. On your labelled set (Step 5), decide whether to threshold this product or the direct question.

Step 5 — Label a small set and choose an automation threshold

Label 50–200 of your own notes for eligibility, then run `experiments/operating_points.py`. It returns the confidence τ that keeps auto-decisions within your error budget, and the share of notes that clears it. Never reuse a τ measured on one model, checklist, facts setting or note source for another.

Step 6 — If coverage is too low, fine-tune on your laptop

Write a seed generator for your checklist (`clinical_system1/vignettes.py`), train on one template family and test on another:

```
upstream/laya/.venv/bin/python experiments/finetune_laya.py --epochs 2 --out runs/laya-clinical
```

Then repeat Step 5. The threshold belongs to the model you ship.

Step 7 — Deploy with review

The local web app (`app/run.sh`, Figure 3) takes a note or a batch, a checklist (built-in or edited in the browser), a model and an error budget. It returns per-criterion probabilities, $P(\text{eligible})$, the audit trail, and an *auto-eligible / auto-ineligible / clinician review* decision. For real use:

- add PHI redaction and audit logging (both upstreams provide prediction hooks);
- have a clinician review a sample of automated negatives;
- monitor coverage and error on a rolling labelled sample.

B Question wording

Lung cancer screening with low-dose CT (USPSTF 2021 (Grade B), JAMA 2021;325(10):962-970).

- `lcs_age` (required: yes): Is the patient between 50 and 80 years old, inclusive?
- `lcs_pack_years` (required: yes): Does the patient have a smoking history of at least 20 pack-years (packs per day multiplied by years smoked; 20 cigarettes = 1 pack)?

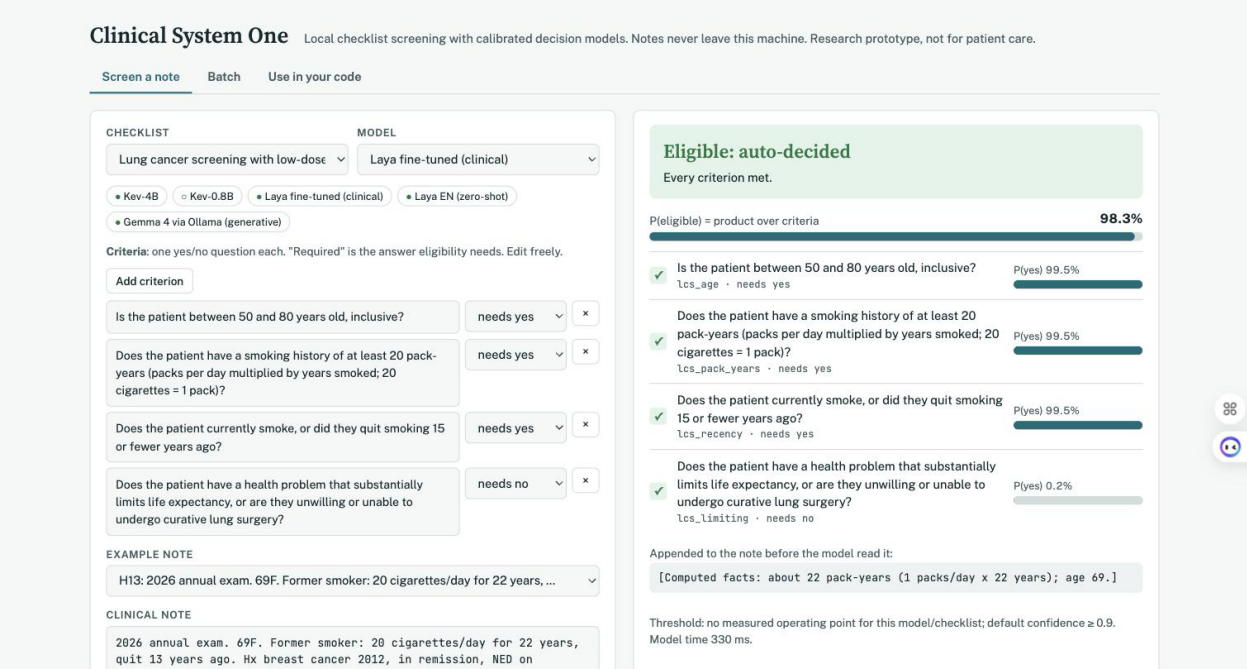


Figure 3: The released local screening app. Note, checklist and model are on the left; per-criterion $P(\text{yes})$, the conjunctive $P(\text{eligible})$, appended computed facts and the automation decision at the chosen error budget are on the right.

- `lcs_recency` (required: yes): Does the patient currently smoke, or did they quit smoking 15 or fewer years ago?
- `lcs_limiting` (required: no): Does the patient have a health problem that substantially limits life expectancy, or are they unwilling or unable to undergo curative lung surgery?
- `lcs_eligible` (direct): Under the USPSTF 2021 lung cancer screening recommendation (adults aged 50 to 80 years who have a 20 pack-year smoking history and currently smoke or have quit within the past 15 years, and who do not have a health problem that substantially limits life expectancy or the ability or willingness to have curative lung surgery), is this patient eligible for annual low-dose CT screening?

One-time abdominal aortic aneurysm ultrasound screening (USPSTF 2019 (Grade B), JAMA 2019;322(22):2211-2218).

- `aaa_male` (required: yes): Is the patient male?
- `aaa_age` (required: yes): Is the patient between 65 and 75 years old, inclusive?
- `aaa_ever_smoked` (required: yes): Has the patient ever smoked tobacco (current or former smoker)?
- `aaa_eligible` (direct): Under the USPSTF 2019 recommendation (one-time ultrasound screening for abdominal aortic aneurysm in men aged 65 to 75 years who have ever smoked), is this patient eligible for AAA screening?

TREC CT 2021.

- `eligible`: Based on the patient description, is this patient eligible for the clinical trial, meaning they satisfy the inclusion criteria and meet none of the exclusion criteria?
- `meets_inclusion`: Does the patient satisfy the trial's inclusion criteria?
- `meets_exclusion`: Does the patient meet any of the trial's exclusion criteria?

C Reproducing every number

```
experiments/prep_trec.py            # TREC CT 2021 sample + ClinicalTrials.gov criteria
experiments/run_all.sh              # original systems, one at a time, resumable
experiments/run_facts.sh            # computed-facts ablation
experiments/bench_model.sh          # any local System One server (new models)
experiments/analyze.py              # summary.json
experiments/figures.py; paper/make_tables.py
```